Proposal for a Swiss AI Initiative Small Project

Project Name

An Open, Multilingual, Multi-hop Knowledge Layer for Apertus

Applicants

(names, institutions, and institutional email addresses; underline scientific lead):

- Joel Barmettler, University of Zurich, joel.barmettler@uzh.ch
- Prof. Dr. Abraham Bernstein, University of Zurich, bernstein@ifi.uzh.ch *[to confirm]*
- Dr. Luca Rossetto, [institution], [email] *[to confirm]*

Request Summary

Cost Dimension	Amount
Alps Compute Hours (GPU-h)	6,000 GPU-h
Alps Storage Space (TB)	5 TB
Number of files to store simultaneously	∼300,000

Proposal Description (max. 2 A4 pages)

a. Purpose of the Swiss AI Initiative Small Project

Large language models (LLMs) hold knowledge either *in their weights* (uneditable, and unreliably recalled for long-tail entities [3]) or *as retrieved text* (faithful but costing hundreds of context tokens per query). We develop a third channel: compressing a knowledge-graph entity's neighborhood into $k \approx 8$ continuous *concept tokens* consumed natively by a *frozen* LLM — attributable (each token traces to citable Wikidata edges), editable, and ∼10× cheaper than text at equal knowledge. Our published predecessor (WWW Companion '26, **best paper award** of its hosting workshop [1]), and the modern rebuild we describe below, establish that the approach works. The purpose of this Small Project is to turn a validated single-graph, single-language, single-hop method into an **open, multilingual, multi-hop knowledge layer for the Apertus mini models** [2] — four scale-ups we have each de-risked with a first result but cannot complete on consumer hardware.

b. Current Situation of the Thematic Area

Two literatures approach the problem. *Soft-token compression* (gist tokens, ICAE, PISCO, xRAG [4, 5, 6, 7]) compresses *text* into continuous tokens; xRAG is closest to us (frozen decoder, same-model distillation) but translates a single retriever vector rather than learning compression, and evaluates neither unseen-entity generalization nor faithfulness. *Knowledge-graph injection* (GNP, GraphToken, KBLaM, Knowledge Prompts [8, 9, 10, 11]) encodes graphs per question with labeled QA, or pre-computes per-entity vectors as lookup tables that cannot handle unseen entities; 2026 meta-analyses find these graph-token systems are not faithful carriers of structure [12]. No prior work combines an *amortized, inductive* per-entity encoder, *label-free* same-model distillation, and *causal* faithfulness evaluation.

We rebuilt the method on modern instruction-tuned backbones and hardened the evaluation (full 14,266-question PopQA on *unseen* entities, leakage-free held-out, Wilson CIs, paired McNemar, 3 seeds). Established, fully reproducible from the public repo and W&B: (i) eight concept tokens lift a frozen Qwen3-0.6B from 0.10 to 0.48 exact-match on unseen entities, where text baselines need ∼5–6× more tokens for parity, and an untrained-injection control (0.13) confirms the trained encoder is the effect; (ii) unseen-entity accuracy scales 2.05× from 10k to 100k training entities *without saturating*; (iii) counterfactual edge-swap probes show the model genuinely *reads* the injected graph; (iv) the recipe transfers from Qwen3 to Gemma-3 *without retuning*, and a Wikidata-trained encoder transfers zero-shot to two unseen graphs (movies, football). Three limits, hit on 24 GB consumer GPUs, define this project: corpora beyond 100k entities exceed our throughput while the data axis is still rising; an English-trained encoder does not carry across the concept *label language* (German-labeled concepts sit near the no-injection floor, ∼9% vs 42–50% for English), a boundary we located precisely; and our concept tokens compress only a 1-hop neighborhood, with multi-hop composition validated so far only as a proof of concept.

c. Activities

All work runs on the **Apertus mini family** [2] (Apertus-v1.1 Instruct at 0.5B, 1.5B, 4B; primary backbone 0.5B, with Qwen3 and Gemma-3 as validated controls), open Wikidata-derived corpora, and our public, CI-gated pipeline. Four work packages, each extending a first result from §b: **WP1 — 1M entities:** complete the data-scaling law from 100k to 300k (partly built) and 1M, to locate where returns level off and whether 1M closes the gap to text-RAG. **WP2 — Multilingual:** train the encoder on target-language Wikidata labels across a representative set of Apertus's languages, measuring per-language unseen-entity accuracy and answer-language fidelity. **WP3 — Multi-hop:** represent each neighbor by its *own* concept vector so a shared-weight encoder composes across hops; build the Wikidata multi-hop corpus and train the recursion label-free, turning the neighborhood compressor into a learned graph embedder. **WP4 — Apertus mini backbones:** apply the full pipeline across 0.5B/1.5B/4B for the first knowledge-injection results on the Swiss open model, and the marquee combination — a multilingual, multi-hop concept layer on Apertus.

Compute is justified from a *measured* anchor: one converged run (0.6B backbone, 100k entities) takes 6.0 wall-hours on an RTX 4090 (public W&B logs); at a conservative 2× GH200 speedup and 1.4× convergence margin, a run costs $4.2 \cdot P \cdot D$ GPU-h (P = active-parameter ratio vs. 0.8B, D = entity ratio vs. 100k). Budget (GPU-h): **WP1** 1M-corpus generation and prep + k-family × 3 seeds × {300k, 1M} (880); **WP2** per-language QA generation, teacher extraction, and multilingual + monolingual training (344); **WP3** 2-hop corpus + recursive hop-2/hop-3 training (516); **WP4** the Apertus 0.5B/1.5B/4B sweep ($\sum P = 7.5$) plus multilingual and 1M corners (742); cross-cutting evaluation over ∼120 checkpoints (150). Subtotal 2,632; +25% contingency (restarts, cluster onboarding) = 3,290; a named stretch tier (a 3M-entity point, the full multilingual × multi-hop × Apertus corner, additional languages, convergence verification) adds ∼1,840 ⇒ **request 6,000 GPU-h**; a ∼2,600 fallback preserves all four WPs' core results. The workload is ∼120 independent single-node jobs; what our own hardware cannot sustain is corpus generation at 1M entities and per-language teacher extraction, not model size. Storage (corpora, teacher caches, ∼120 checkpoints, per-item dumps) totals ∼3.5 TB; with headroom we request 5 TB (∼300k files).

d. Team

The team carried this line of work through peer review once already: v1 [1] (Barmettler, Bernstein, Rossetto) won its hosting workshop's best paper award at The Web Conference 2026. J. Barmettler (scientific lead, PhD candidate UZH) built the v2 system end-to-end: data pipeline, training, the hardened evaluation protocol, and the cross-family and cross-lingual studies. Prof. A. Bernstein (UZH) contributes expertise in knowledge graphs and the Semantic Web; Dr. L. Rossetto contributes expertise in multimodal retrieval and evaluation [roles to confirm]. During the funding period we run weekly reviews against the pre-registered plan and share W&B dashboards with per-item dumps for independent verification of every claim. [Multi-node/Alps experience to be added.]

e. Expected Outputs and Outcomes

Within six months: (1) a **paper** with the first data-scaling law for an inductive knowledge encoder, the multilingual and multi-hop results, and the causal-faithfulness methodology; (2) **open data** — 300k/1M, multilingual, and multi-hop Wikidata corpora (CC0, sha-manifested); (3) **open weights** — all encoders, including Apertus-mini concept layers, on HuggingFace; (4) **open source** — the full CI-gated pipeline and eval harness (already public); (5) the **open results dataset** for GPU-free re-analysis. The project delivers a provenance-carrying knowledge layer purpose-built for Apertus and seeds a **Large Project** that trains it at the design points these results identify.

f. Importance for Switzerland, Europe, and beyond

The project is a direct fit for the Initiative's own model. ConceptFormer's only knowledge source is **Wikidata (CC0)**, training is **label-free** (no proprietary QA), and all code, weights, corpora, and per-item outputs are released — the same open-data, open-weights, open-recipe stance on which Apertus [2] was built, and WP2 amplifies its flagship multilingual capability. A knowledge channel that grounds open models in curated, auditable sources (every concept token traces to citable KG edges, and our counterfactual protocol proves the model uses them), rather than opaque weights, is a building block for European digital sovereignty: deployments can maintain their own knowledge graphs and update model knowledge without retraining or per-query retrieval, with the whole value chain reproducible.

g. Ethics and Regulatory Compliance

Data: Wikidata (CC0) and public benchmarks; no personal data beyond encyclopedic facts about public entities, no human subjects, no scraping. Generated training questions come from open-weight models and are released with their prompts; all artifacts carry explicit licenses. The approach is transparency-*enhancing*, aligned with EU AI Act expectations; we comply with UZH research-integrity and CSCS usage policies and publish all results, including negative ones.

Signature

By submitting this proposal, we confirm that all leading applicants have understood the requirements for participation according to the call document.

Place(s) and Date(s):

Names and signatures of the leading applicants:

References

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